Yujian Xiong

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Education

Arizona State University *Tempe, AZ*
Ph.D., Data Science, Analytics and Engineering *Jan 2022 – Jan 2027*

- **Advisor:** Prof. Yalin Wang. **Co-Advisor:** Prof. Zhong-Lin Lu (**NYU**). **GPA:** 4.0/4.0

University of California San Diego *La Jolla, CA*
M.S., Machine Learning and Data Science *Sep 2019 – Mar 2021*

- **Advisor:** Prof. Duygu Kuzum. **GPA:** 3.9/4.0

Yale University *New Haven, CT*
Exchange Scholar, Department of Cell Biology *Feb 2018 – Jun 2018*

Fudan University *Shanghai, China*
B.S., Physics *Sep 2014 – Jun 2019*

Game Projects

Small-Team Indie UE5 Platformer Hacker Game Development *Nov 2025 – Present*

- Architected core gameplay in UE5 using C++ and Blueprint for rapid prototyping and iterations.
- Integrated an API-based conversation system into the game loop, for dynamic quest dialogue and context-aware hacking interactions driven by commercial LLM inference.

Individual Indie RPG Development *Dec 2023 – Present*

- Designed tabletop-RPG style statistic systems, including turn-based combat and multi-class character builds.
- Trained personalized diffusion models for pixel-style character art, 2D dungeon map, and VFX design.
- Constructed and coded the software using Godot4.

Tabletop-RPG Design and Localization *Jun 2022 – Nov 2023*

- Main contributor to the Chinese localization of a JRPG style rule "Fabula Ultima" from Italian.
- Helped the localized version towards crowdfunding, reached highest history in Chinese TRPGs (¥2.6M).

Research Projects

Ophthalmology Foundation VLLM *ASU, AZ* *Dec 2025 – Present*

- Curated large-scale medical vision-language dataset of 1M entries from 15k hours Youtube videos via an automated pipeline integrating DINOv3-based frame mining, Whisper ASR, Qwen-based densification.
- Trained CLIP alignment via contrastive pretraining, achieving sota zero-shot cross-modal retrieval against strongest medical baseline (BMC-CLIP).
- Built a multimodal clinical reasoning agent powered by open-source Qwen to enable image-guided diagnosis and structured ophthalmic report generation without reliance on private commercial models.

Optimization and ViT Innovation for Multimodal LLMs *ASU, AZ* *Aug 2025 – Present*

- Developed a Large Language Diffusion Model for medical VQA, replacing autoregressive decoding with masked diffusion to enhance report generation and logical consistency.
- Engineered a training-free pruning framework using Optimal Transport and Wasserstein distance to mitigate ViT token redundancy, significantly reducing inference FLOPs without accuracy loss.

Transformer-based Octree MAE Pipeline for Tetrahedral and Triangular 3D Meshes *ASU, AZ* *Jul 2024 – Present*

- Designed an MAE with octree-guided transformer attention for large tetrahedral/triangular 3D meshes.
- Use simplex-aware octree partitioning and positional encoding to enable scalable learning across variable mesh resolutions/sizes.
- Unified multi-source mesh information besides coordinates (geometry/attributes) within the unsupervised

MAE framework, supporting both medical brain MRI mesh and general 3D mesh tasks.

3T to 7T fMRI Time Series Signal Enhancement and Denoising Using GAN and Bridge-based Diffusion Models *ASU, AZ* *Oct 2023 – Present*

- Built time series fMRI signal processing pipeline upon 3D/2D brain surface meshes.
- Developed GAN & Schrödinger bridge Diffusion frameworks to enhance low-quality fMRI on brain mesh.
- Achieved downstream improvements on low-quality (3T) fMRI on tasks including pRF modeling, visual image reconstruction, demonstrating fidelity comparable to native high-quality (7T).

Geometry fMRI Analysis and Processing for Human Vision *ASU, AZ* *Jan 2022 – Present*

- Advanced geometry processing including conformal mapping, topology correction, and optimal transport for fMRI signal on 3D/2D brain surface meshes.
- Streamlined ML pipelines for multiple 3T/7T fMRI datasets, including DNNs/transformers/diffusions.
- Leveraged geometry & ML methods to study visual processing through fMRI (pRF, cortical magnification...)

Work Experience

Machine Learning Researcher Intern *Shanghai, China*
Netease Interactive Entertainment *May 2021 – Aug 2021*

- Researched and implemented a chatbot system utilizing GPT-2 for Mandarin RPG narration.
- Investigated LLM-based narration using local finetuning and prompt engineering with knowledge graphs.
- Delivered a functional Unity-based demo with Dockerized deployment.

Machine Learning Engineer Intern *Shanghai, China*
Cozy-Cat Technology Co. Ltd *Aug 2021 – Dec 2021*

- Calibrated and optimized a multi-camera markerless motion detection system; analyzed data and recordings.
- Modified 2021 SOTA models (OpenPose, MoveNet), developed a 3D pose estimation network leveraging point-cloud and multi-view recordings.
- Maintained the detection framework and Docker interface for real-time latency with Unity apps.

Research Publications

OctEncoder: Octree-based Transformer MAE on Tetrahedral and Triangular Meshes 2026

Yujian Xiong[1], Mohammad Farazi[1], Xuanzhao Dong, Sebastian Waz, Zhong-Lin Lu*, Yalin Wang*

MICCAI 2026 (Under review)

RFT: Receptive Field Transformer for Human Brain Visual fMRI Response Prediction 2026

Yujian Xiong, Sebastian Waz, Xuanzhao Dong, Wenhui Zhu, Xin Li, Zhong-lin Lu*, Yalin Wang*

ECCV 2026 (Under review)

AHA: Aligning Large Audio-Language Models for Reasoning Hallucinations via Counterfactual Hard Negatives 2026

Yanxi Chen, Wenhui Zhu, Xiwen Chen, Zhipeng Wang, Xin Li, Peijie Qiu, Hao Wang, Xuanzhao Dong, ***Yujian Xiong***, Anderson Schneider, Yuriy Nevmyvaka, Yalin Wang*

ACL 2026 Findings Workshop arXiv:2512.24052

OTPrune: Distribution-Aligned Visual Token Pruning via Optimal Transport 2026

Xiwen Chen, Wenhui Zhu, Gen Li, Xuanzhao Dong, ***Yujian Xiong***, Hao Wang, Peijie Qiu, Qingquan Song, Zhipeng Wang, Shao Tang, Yalin Wang*, Abolfazl Razi*

CVPR 2026 Findings Workshop arXiv:2602.20205

SGW-GAN: Sliced Gromov-Wasserstein GANs for Retinal Fundus Image Enhancement 2026

Yujian Xiong, Xuanzhao Dong, Wenhui Zhu, Xin Li, Oana Dumitrascu*, Yalin Wang*

WACV 2026 P2P-CV **Oral**. arXiv:2601.13417

nnMobileNet++: Towards Efficient Hybrid Networks for Retinal Image Analysis 2025

Xin Li, Wenhui Zhu, Xuanzhao Dong, Hao Wang, ***Yujian Xiong***, Oana Dumitrascu*, Yalin Wang*

WACV 2026 P2P-CV.

Bridging Restoration and Diagnosis: A Comprehensive Benchmark for Retinal Fundus Enhancement 2025

Xuanzhao Dong, Wenhui Zhu, Xiwen Chen, Hao Wang, Xin Li, ***Yujian Xiong***, Jiajun Cheng, Zhipeng Wang, Shao Tang, Oana Dumitrascu*, Yalin Wang*

WACV 2026 P2P-CV.

VAOT: Vessel-Aware Optimal Transport for Retinal Fundus Enhancement 2025

Xuanzhao Dong, Wenhui Zhu, ***Yujian Xiong***, Xiwen Chen, Hao Wang, Xin Li, Jiajun Cheng, Zhipeng Wang, Shao Tang, Oana Dumitrascu*, Yalin Wang*

WACV 2026 P2P-CV **Oral**. arXiv:2511.18763

Quantification of Planar Cortical Magnification with Optimal Transport and Topological Smoothing 2025

Yujian Xiong, Negar Jalili Mallak, Yanshuai Tu, Zhong-Lin Lu*, Yalin Wang*

NeuroImage. doi:10.1016/neuroimage.2026.121690

Reconstructing Retinal Visual Images from 3T fMRI Data Enhanced by Unsupervised Learning 2024

Yujian Xiong, Wenhui Zhu, Zhong-Lin Lu*, Yalin Wang*

ISBI 2024 **Oral**. arXiv:2404.05107

DME-MobileNet: Fine-Tuning nnMobileNet for Diabetic Macular Edema Classification 2024

Xuanzhao Dong, Yalin Wang*, Yanxi Chen, Xin Li, Hao Wang, Peijie Qiu, Xiwen Chen, Abolfazi Raze, ***Yujian Xiong***, Oana Dumitrascu, Wenhui Zhu

MICCAI Challenge UWF4DR 2024. Paper Page

CUNSB-RFIE: Context-aware Unpaired Neural Schrödinger Bridge in Retinal Fundus Image Enhancement 2024

Xuanzhao Dong, Vamsi Krishna Vasa, Wenhui Zhu, Peijie Qiu, Xiwen Chen, Yi Su, ***Yujian Xiong***, Zhangsihao Yang, Yanxi Chen, Yalin Wang*

WACV 2024. arXiv:2409.10966

Visual Inputs Reconstructing through Enhanced 3T fMRI Data from Optimal Transport Guided Generative Adversarial Network 2024

Yujian Xiong, Wenhui Zhu, Zhong-Lin Lu*, Yalin Wang*

Journal of Vision 24(10):1478–1478

Context-Aware Optimal Transport Learning for Retinal Fundus Image Enhancement 2024

Vamsi Krishna Vasa, Peijie Qiu, Wenhui Zhu, ***Yujian Xiong***, Oana Dumitrascu, Yalin Wang*

WACV 2024. arXiv:2409.07862

Characterizing visual cortical magnification with topological smoothing and optimal transportation 2023

Yujian Xiong, Yanshuai Tu, Zhong-Lin Lu*, Yalin Wang*

SPIE Medical Imaging 2023. arXiv:2404.06505

Planar Cortical Magnification Measurement with Topological Smoothing and Optimal Transportation 2022

Yujian Xiong, Yanshuai Tu, Zhong-Lin Lu*, Yalin Wang*

12th Annual ASU–BNI Neuroscience Research Symposium

Sec1/Munc18-Family Proteins Catalyze Directional SNARE Assembly by Templating SNARE Folding and Association 2019

Junyi Jiao, Mengze He, Sarah Port, Baker Richard, Yonggang Xu, Hong Qu, ***Yujian Xiong***, Yukun Wang, Huaizhou Jin, Travis Eisemann, Frederick M. Hughson, Yongli Zhang*

eLife. doi:10.7554/eLife.41771

Technologies

Languages: Python, MATLAB, C++, C#, SQL, JavaScript, GDScript

Technologies: MATLAB, PyTorch, TensorFlow, Blender, Freesurfer, LabVIEW, .NET, Xcode